Summary

LAB 2 | MINDBRIDGE APPLICATION REVIEW

Application	MindBridge
Environment	http://localhost:3000
Inspection tool	Google Chrome DevTools — Lighthouse

1. Overview

The review identified two accessibility violations and two usability opportunities. The accessibility findings relate to missing alternative text on course images and insufficient text contrast. The usability observations concern clarity in the registration wizard and recovery guidance when registration fails.

2. Accessibility findings

Finding 1 — Missing alternative text on course images

Defect ID	D-WCAG-ALT
Affected elements	Course Detail hero image (course-hero) and Home/Catalogue course thumbnails (course-thumb)
WCAG criterion	1.1.1 — Non-text Content
Conformance level	Level A
Inspection method	Google Chrome DevTools → Lighthouse → Accessibility audit; inspect flagged images in the Elements panel

Expected behavior: Meaningful course images should include descriptive alternative text, such as “Cover image for [course title]”, so assistive technologies can communicate the image’s purpose.

Observed behavior: Course images on the Home, Catalogue, and Course Detail pages render without an alt attribute.

User impact: Screen-reader users may hear the images announced as unlabeled or generic content and cannot determine which course each image represents. This makes it harder to browse and select courses independently.

Recommended action: Add concise, context-appropriate alt text to meaningful course images. If an image is purely decorative and conveys no information, use an empty alternative (alt="") rather than omitting the attribute.

Finding 2 — Insufficient contrast on primary call-to-action

Defect ID	D-WCAG-CONTRAST

Primary affected element	Home page “Browse courses” button (hero-cta)
Secondary affected elements	Featured course metadata (featured-meta), including instructor (featured-instructor) and price (featured-price)
WCAG criterion	1.4.3 — Contrast (Minimum)
Conformance level	Level AA
Inspection method	Google Chrome DevTools → Lighthouse → Accessibility audit; inspect contrast issues using the Elements panel color picker

Expected behavior: Normal-sized text should have a contrast ratio of at least 4.5:1 against its background. Large text has a minimum requirement of 3:1.

Observed behavior: The “Browse courses” button uses indigo text on an indigo background. The lab findings record a contrast ratio below 2:1. The same low-contrast treatment appears in the featured course instructor and price details.

User impact: People with low vision or colour-perception differences, as well as users viewing the page in bright ambient light, may have difficulty reading the main call-to-action. The low contrast also makes featured-course information difficult to read.

Recommended action: Change the text or background colour so the measured contrast meets WCAG AA requirements. Recheck the button label and the featured-course metadata after the styling change.

3. Usability observations and recommendations

Observation 1 — Registration wizard step clarity

Issue: The three-step registration wizard (Email → Password → Profile) displays step numbers but does not explain until Step 3 which fields are required and which are optional.

Recommendation: Add a clear note on Step 1 explaining that all three steps are required to create an account. On Step 3, consistently mark optional fields so users can tell which information may be skipped.

Expected benefit: Users understand the process and its requirements before they begin, reducing uncertainty and avoidable interruptions.

Observation 2 — Registration error recovery

Issue: When registration fails because of a server error, the Step 3 message is generic and does not explain what the user should do next.

Recommendation: Provide specific, actionable messages. For a duplicate email address, display: “This email is already registered — sign in instead.” For an unexpected server error, display: “Something went wrong — please wait a moment and try again.”

Expected benefit: Clear recovery guidance helps users understand the problem and choose an appropriate next step.

4. Summary

This accessibility and usability review of the MindBridge application (http://localhost:3000), using Google Chrome DevTools — Lighthouse, identified two accessibility violations and a usability improvement. First, course images on the Home, Catalogue, and Course Detail pages have no alternative text, failing WCAG 1.1.1, Non-text Content (Level A). Without descriptive alternatives, screen-reader users cannot identify the courses represented by the images, making independent browsing more difficult. Second, the Home page's "Browse courses" call-to-action uses indigo text on an indigo background. The measured contrast ratio is below 2:1, which does not meet the 4.5:1 minimum for normal text under WCAG 1.4.3, Contrast (Minimum) (Level AA). This may prevent people with low vision, colour-perception differences, or users viewing the page in bright light from reading the primary action. The same low-contrast styling affects featured-course instructor and price details. As a usability improvement, the registration wizard should explain on Step 1 that all three steps are required and consistently identify optional fields on Step 3. This would set expectations and help users complete registration with less uncertainty.

5. Conclusion

The review identified two accessibility issues that should be prioritized: adding appropriate alternative text to meaningful course images and correcting text contrast on the primary call-to-action and featured-course metadata. The registration wizard would also benefit from clearer information about required steps and more useful error-recovery messages. Addressing these items would improve access to course information and make key registration and browsing tasks easier to understand.